PROJECT MEATLENS: DEVELOPMENT AND EVALUATION OF A CNN-BASED PORK MEAT FRESHNESS CLASSIFICATION SYSTEM FOR IMAGE GUIDED DECISION SUPPORT IN THE IKOT PALENGKE PROGRAM OF OLONGAPO WET MARKETS

Project Leader

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REVISIONS

Scope & Regulatory Compliance

- Tightened project delimitations by strictly defining scannable specimens to prevent scope creep.
- Integrated regulatory compliance measures, including DTI checklists (light, MIC, sanitation) and updated Terms of Service.

Machine Learning & Technical Optimization

- Retrained the machine learning model, improving result accuracy, and added a developer dashboard for ongoing training.
- Bypassed web-app camera limitations by routing to the native phone camera and implementing a blur-prevention algorithm to ensure high-quality specimen captures.

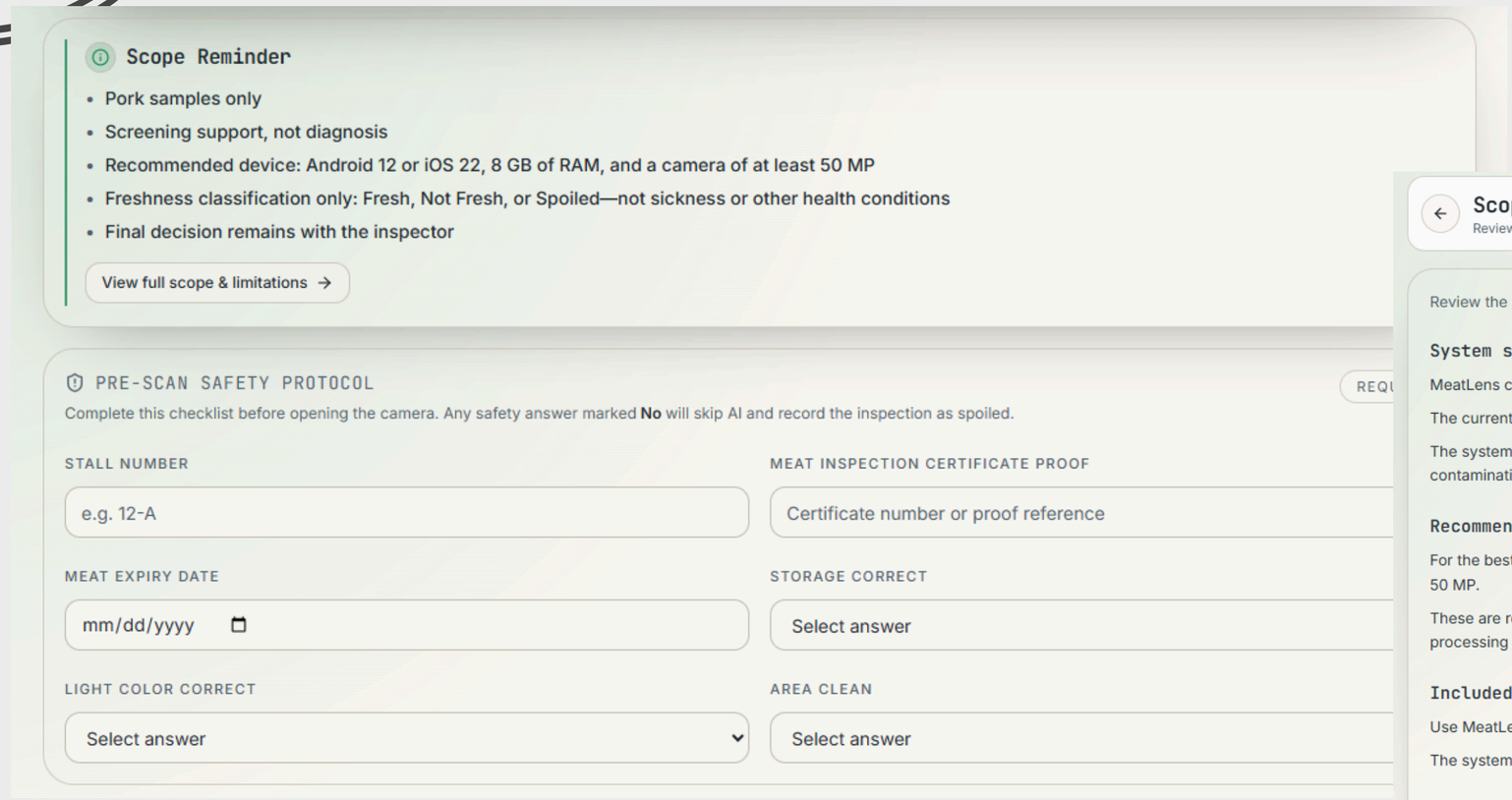
Core Feature Additions

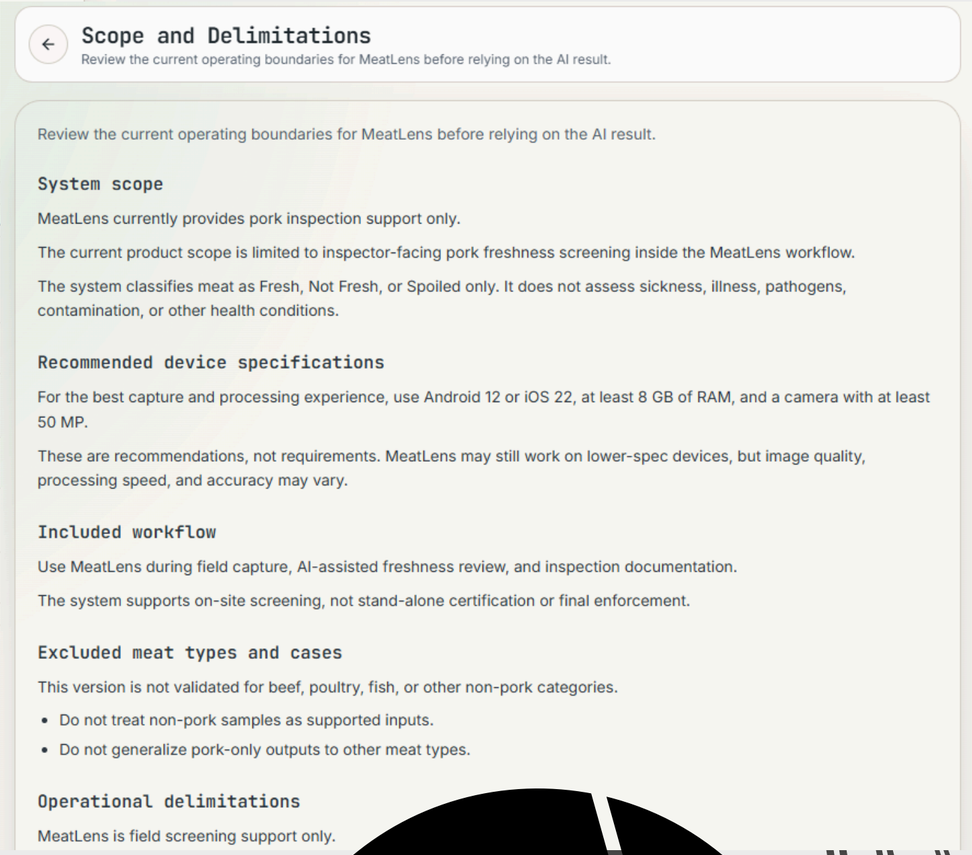
- Introduced GPS location tracking as a redundancy measure alongside manual selection.
- Enabled automated, role-based report generation tailored for Admins and Meat Inspectors.

User Experience & Expert Validation

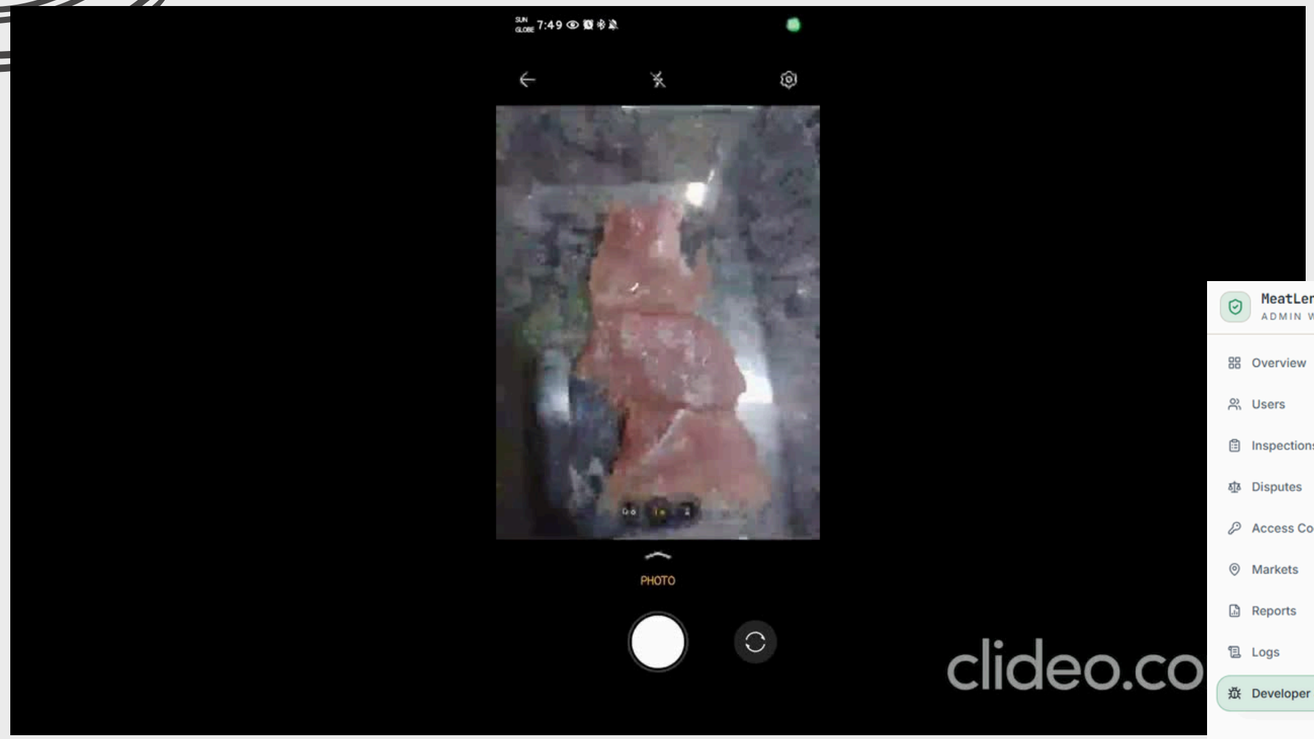
- Implemented mandatory, revisitable in-app user tutorials for first-time onboarding.
- Standardized UI/UX design across tabs, including terminology updates.
- Consulted with veterinary experts and field inspectors to definitively verify meat classification levels.

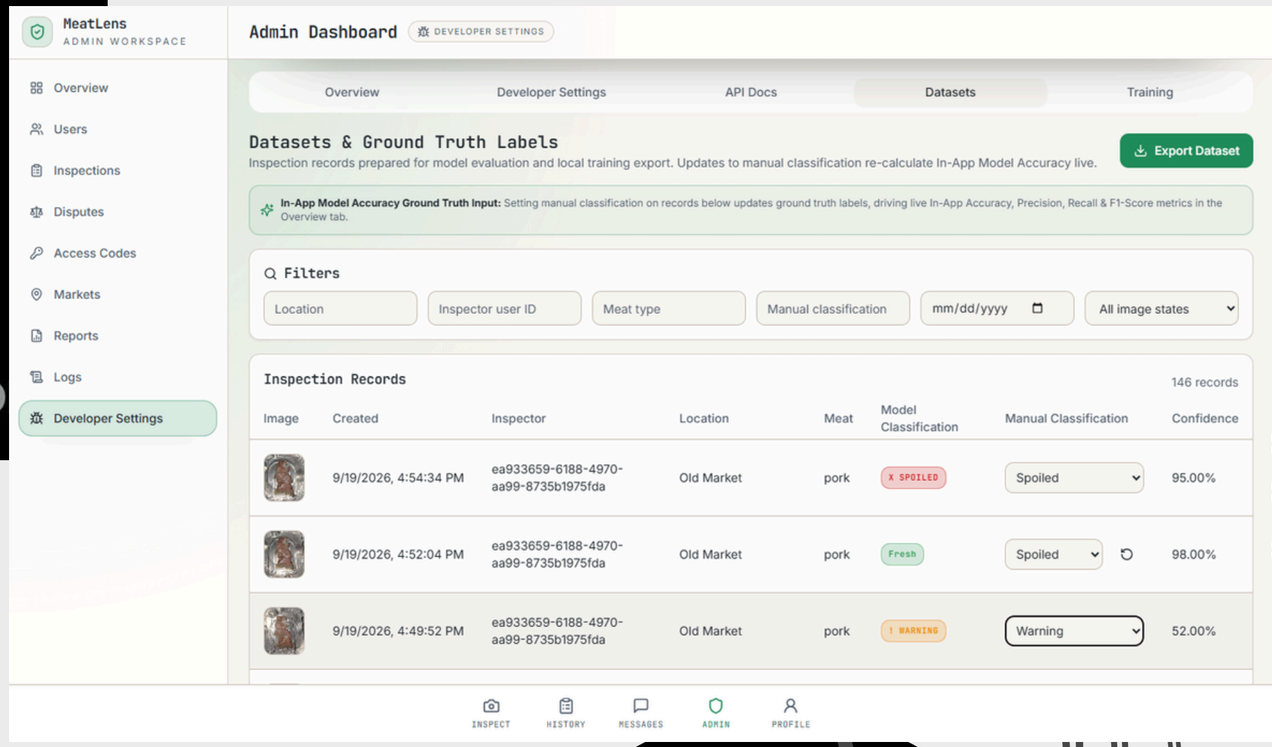
SCOPE & REGULATORY COMPLIANCE





MACHINE LEARNING & TECHNICAL OPTIMIZATION





CORE FEATURE ADDITIONS

Republic of the Philippines
City Government of Olongapo
CITY VETERINARY OFFICE
Eucalyptus St, Gordon Heights, Olongapo City, Zambales 2200

Administrative Report

Range: 2026-08-21 to 2026-09-19

Generated: Sep 19, 2026 9:26 PM

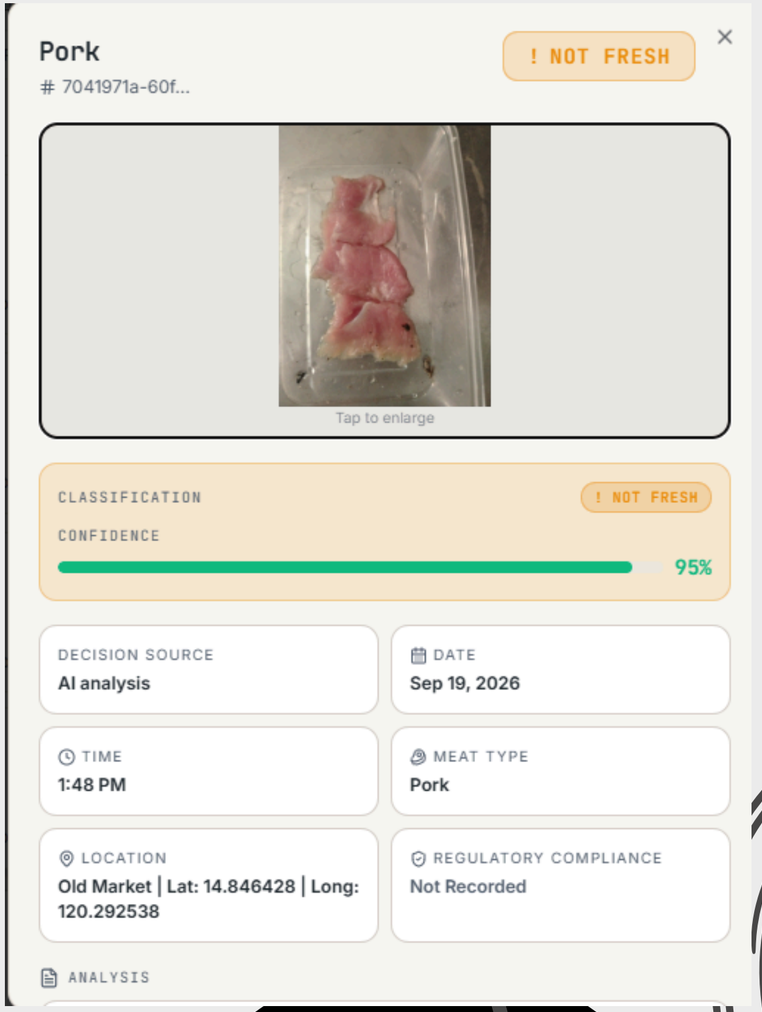
Organization: City Veterinary Office of Olongapo

Veterinary and Meat Safety Overview

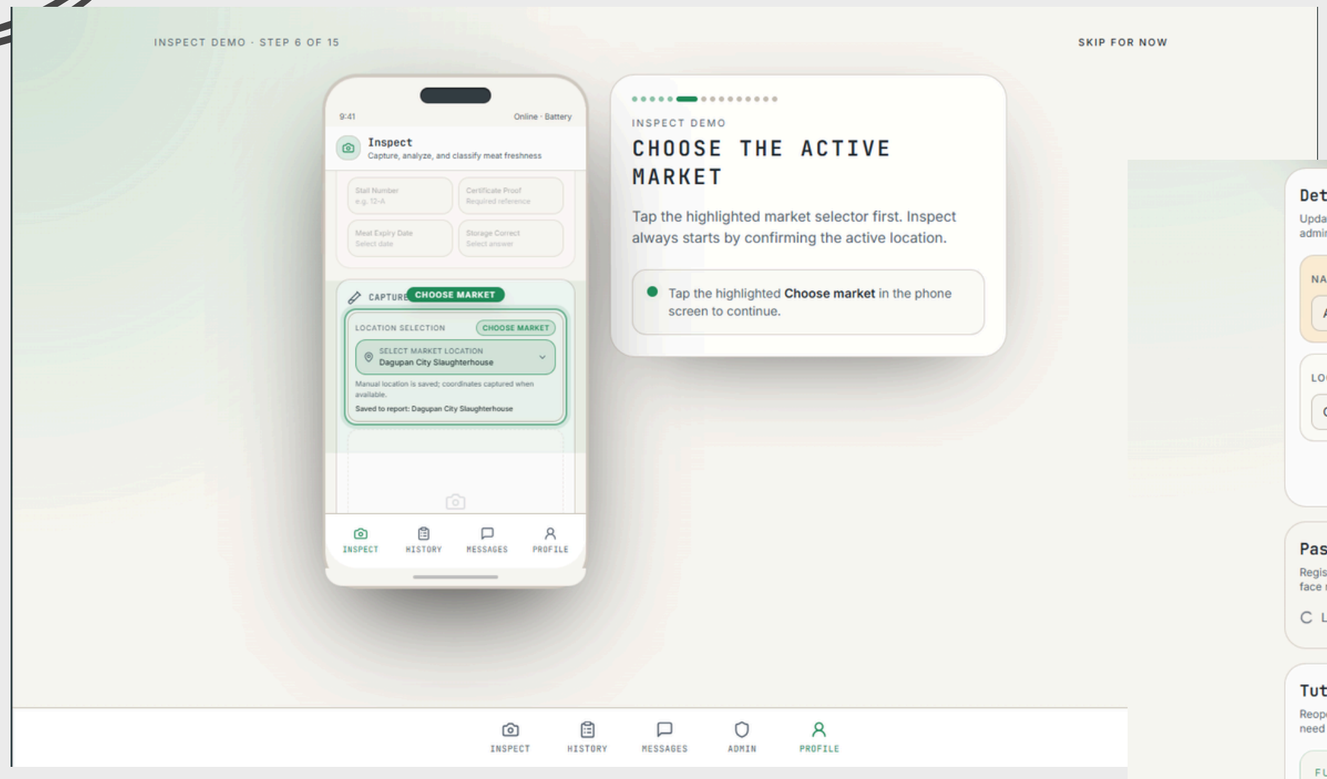
Total Inspections 32	Average Confidence 95%
Spoiled Rate 50%	Unique Inspectors 1
Unique Locations 4	Records With Deviations 0

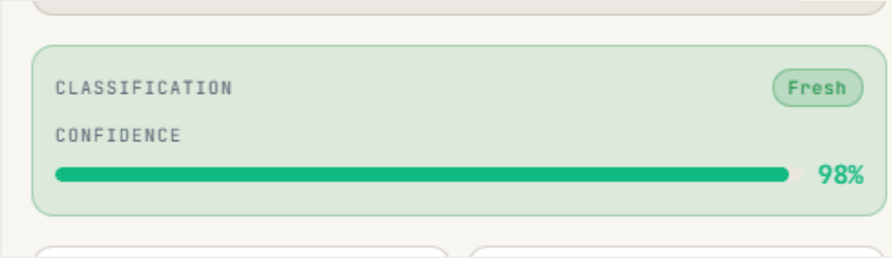
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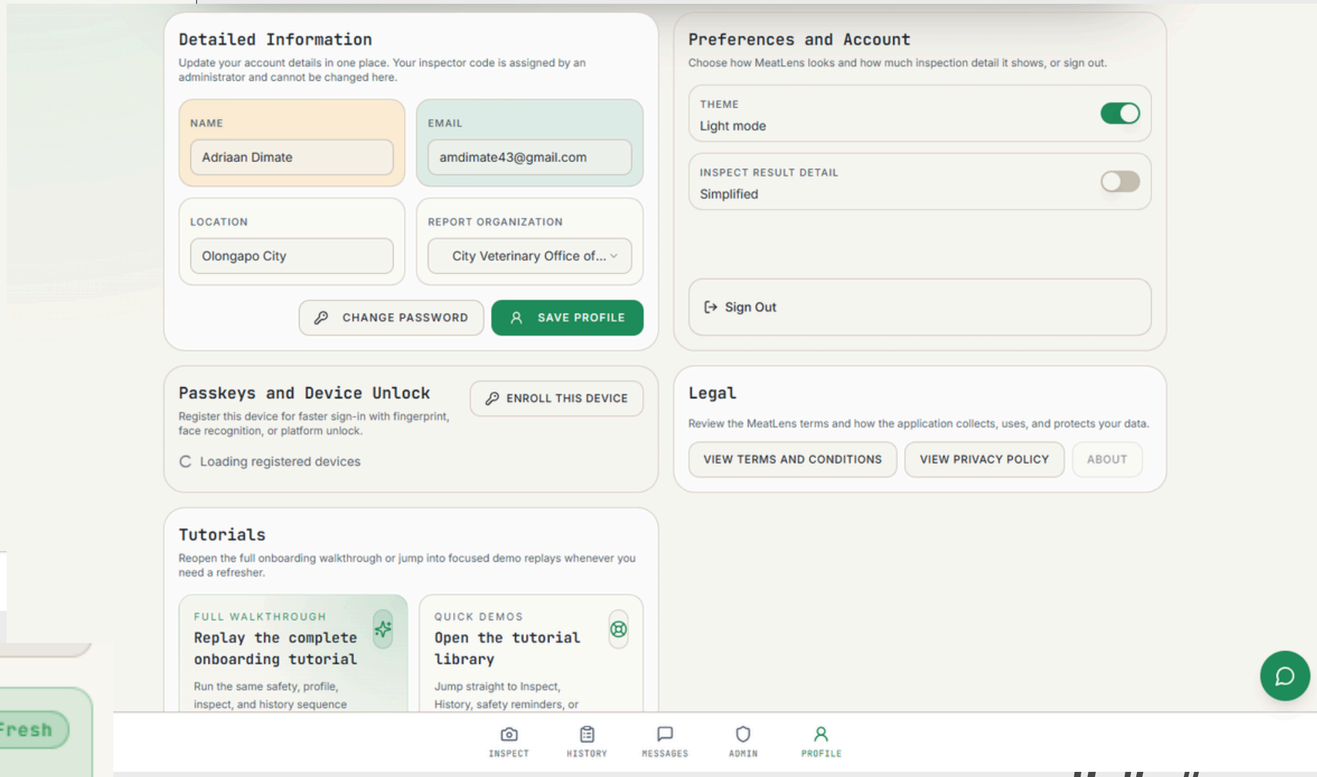
Veterinary Inspection Graphs



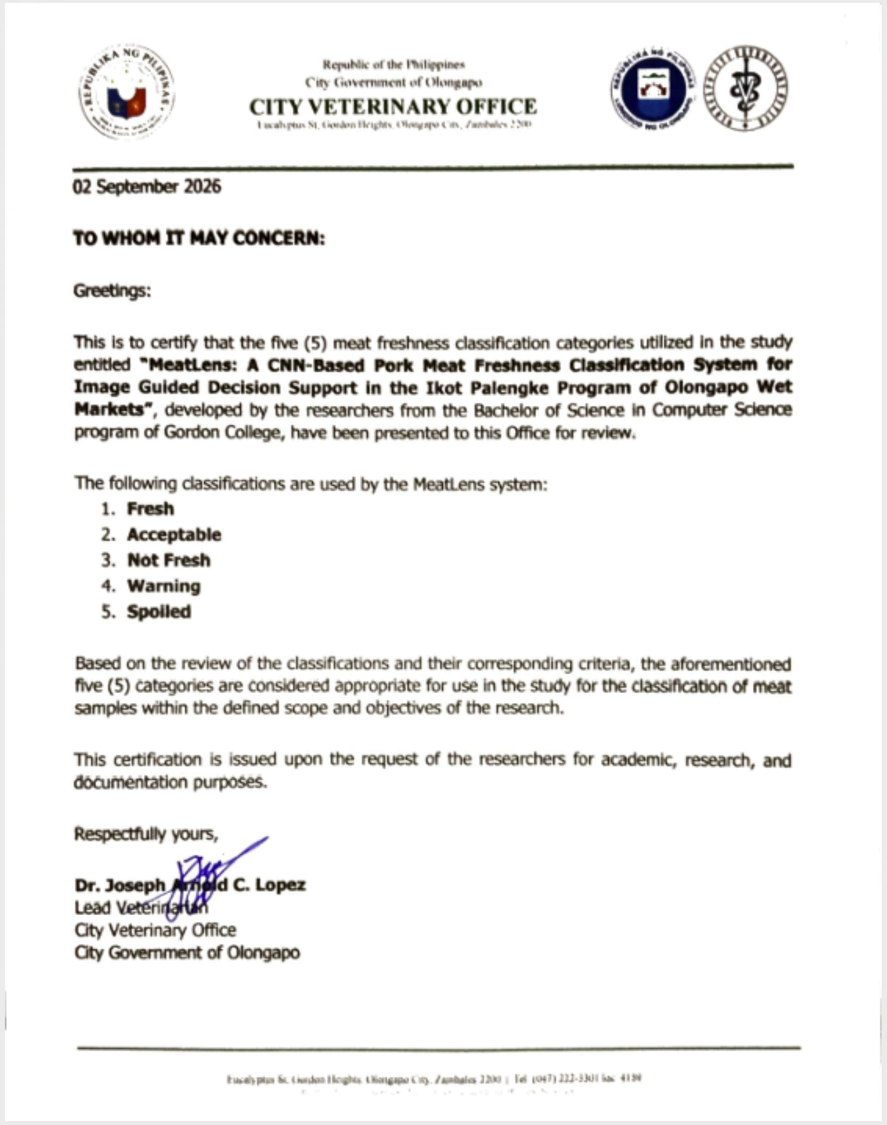
USER EXPERIENCE







EXPERT VALIDATION



Republic of the Philippines
City Government of Olongapo
CITY VETERINARY OFFICE
Eucalyptus St. Gordon Heights, Olongapo City, Zambales 2200

02 September 2026

TO WHOM IT MAY CONCERN:

Greetings:

This is to certify that the five (5) meat freshness classification categories utilized in the study entitled **"MeatLens: A CNN-Based Pork Meat Freshness Classification System for Image Guided Decision Support in the Ikot Palengke Program of Olongapo Wet Markets"**, developed by the researchers from the Bachelor of Science in Computer Science program of Gordon College, have been presented to this Office for review.

The following classifications are used by the MeatLens system:

1. **Fresh**
2. **Acceptable**
3. **Not Fresh**
4. **Warning**
5. **Spoiled**

Based on the review of the classifications and their corresponding criteria, the aforementioned five (5) categories are considered appropriate for use in the study for the classification of meat samples within the defined scope and objectives of the research.

This certification is issued upon the request of the researchers for academic, research, and documentation purposes.

Respectfully yours,

Dr. Joseph Arnold C. Lopez
Lead Veterinarian
City Veterinary Office
City Government of Olongapo

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LIVE DEMONSTRATION



